DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

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DHCP and DORA Process

Learning Objective

To understand how DHCP automatically provides network configuration to a device and learn the four steps of the DORA process.

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1. What is DHCP?

DHCP stands for **Dynamic Host Configuration Protocol**.

It is used to automatically provide network configuration to devices when they connect to a network.

Instead of manually entering an IP address and other network settings, a DHCP server can provide them automatically.

DHCP can provide information such as:

- IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time

DHCP normally uses:

- **UDP port 67** — DHCP server
- **UDP port 68** — DHCP client

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2. What is DORA?

DORA stands for:

- **D — Discover**
- **O — Offer**
- **R — Request**
- **A — Acknowledge**

These are the four main steps that happen when a device requests an IP address from a DHCP server.

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3. Step 1 — DHCP Discover

When a device joins a network, it may not have an IP address yet.

So it sends a **DHCP Discover** message to find available DHCP servers.

The client basically says:

> "Is there any DHCP server available that can give me an IP address?"

This message is normally broadcast because the client may not know the address of the DHCP server.

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4. Step 2 — DHCP Offer

The DHCP server receives the Discover message and sends back a **DHCP Offer**.

The offer can contain:

- IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time

For example:

IP Address: 192.168.1.20

Subnet Mask: 255.255.255.0
Gateway: 192.168.1.1
DNS: 192.168.1.1

The server is basically offering the client an IP configuration.

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5. Step 3 — DHCP Request

The client receives the offer and decides which offer it wants to use.

It then sends a **DHCP Request** message.

This tells the DHCP server that the client wants to use the offered IP address.

It also lets other DHCP servers know that their offers were not selected.

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6. Step 4 — DHCP Acknowledge

Finally, the DHCP server sends a **DHCP Acknowledgement (ACK)**.

This confirms that the client can use the assigned IP address and other network settings.

The device can now communicate on the network using the assigned configuration.

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7. DORA Flow

The complete process can be remembered like this:

Client
  |
  | DHCP Discover

↓
DHCP Server
│
│ DHCP Offer
↓
Client
│
│ DHCP Request
↓
DHCP Server
│
│ DHCP ACK
↓
Client gets IP configuration

Easy way to remember

Discover → Offer → Request → Acknowledge

D → O → R → A

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8. Example

Suppose I connect my laptop to a Wi-Fi network.

Initially, my laptop may not have an IP address for that network.

The process can look like:

1. Laptop sends **DHCP Discover**.
2. Router/DHCP server sends a **DHCP Offer**.
3. Laptop sends a **DHCP Request**.
4. Router sends **DHCP ACK**.
5. Laptop starts using the assigned IP address.

After this, the laptop can communicate with other devices and access the internet.

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9. Why DHCP is Useful for a SOC Analyst

DHCP logs can be useful during an investigation because they can help connect an IP address to a specific device at a particular time.

For example, suppose a SOC analyst sees suspicious traffic coming from:

192.168.1.20

The analyst may check DHCP logs to find:

- Which device had that IP?
- When was the IP assigned?
- What MAC address was associated with it?
- Which hostname was using it?
- Was the same IP previously assigned to another device?

This can help identify the actual device involved in suspicious network activity.

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10. My Understanding

I understood DHCP as a way for devices to automatically get the network information they need.

The main thing I need to remember is **DORA**:

Discover → Offer → Request → Acknowledge

From a SOC point of view, DHCP logs can also be useful because IP addresses can change over time. Checking DHCP information can help connect an IP address to the correct device during an investigation.

11. Basic DHCP Troubleshooting

If a device is not getting an IP address automatically, I can check a few things to find the problem.

Common checks

- Check whether the network connection is working.
- Check the device's current IP address.
- Check whether the DHCP service/server is running.
- Check whether the DHCP server has available IP addresses.
- Check whether the device is getting an address such as `169.254.x.x`.
- Check DHCP logs for errors or failed requests.
- Renew the DHCP lease and check if the device receives a new IP.

On Linux

I can check the network configuration using:

ip addr

I can check the network interface using:

ip -br link

Depending on the Linux setup, I can also renew the DHCP lease using the system's network management tools.

For example, with `dhclient`:

sudo dhclient -r
sudo dhclient

The first command releases the current DHCP lease and the second one requests a new lease.

Example of a problem

If my system gets an address like:

169.254.x.x

it can indicate that the device was unable to get a proper IP address from the DHCP server and assigned itself a link-local address.

In that situation, I would check the network connection, DHCP server availability and DHCP logs.

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